

Apt Training Resources

1. Write a program to give the following output for the given input

Eg 1: Input: a1b10

Output: abbbbbbbbbb

Eg: 2: Input: b3c6d15

Output: bbbccccccdddddddddddddd

The number varies from 1 to 99.

2. Write a program to sort the elements in odd positions in descending order and elements in ascending order

Eg 1: Input: 13,2 4,15,12,10,5

Output: 13,2,12,10,5,15,4

Eg 2: Input: 1,2,3,4,5,6,7,8,9

Output: 9,2,7,4,5,6,3,8,1

3. Write a program to print the following output for the given input. You can assume the string is of odd length

Eg 1: Input: 12345

Output:

1 5

2 4

3

2 4

1 5

Eg 2: Input: indianisgreat

Output:

```
i               t
n               a
d               e
i               r
a               g
n               s
i
```

```
      i
    n  s
  a    g
i      r
 d    e
n      a
i          t
```

4. Find if a String2 is substring of String1. If it is, return the index of the first occurrence. else return -1.

Eg 1: Input:

String 1: test123string

String 2: 123

Output: 4

Eg 2: Input:

String 1: testing12

String 2: 1234

Output: -1

5. Given two sorted arrays, merge them such that the elements are not repeated

Eg 1: Input:

Array 1: 2,4,5,6,7,9,10,13

Array 2: 2,3,4,5,6,7,8,9,11,15

Output:

Merged array: 2,3,4,5,6,7,8,9,10,11,13,15

6. Using Recursion reverse the string such as

Eg 1: Input: one two three

Output: three two one

Eg 2: Input: I love india

Output: india love I

COMPLEX CODING:

1) Design a Call taxi booking application

-There are n number of taxi's. For simplicity, assume 4. But it should work for any number of taxi's.

-There are 6 points(A,B,C,D,E,F)

-All the points are in a straight line, and each point is 15kms away from the adjacent points.

-It takes 60 mins to travel from one point to another

-Each taxi charges Rs.100 minimum for the first 5 kilometers and Rs.10 for the subsequent kilometers.

-For simplicity, time can be entered as absolute time. Eg: 9hrs, 15hrs etc.

-All taxi's are initially stationed at A.

-When a customer books a Taxi, a free taxi at that point is allocated

-If no free taxi is available at that point, a free taxi at the nearest point is allocated.

-If two taxi's are free at the same point, one with lower earning is allocated

-Note that the taxi only charges the customer from the pickup point to the drop point. Not the distance it travels from an adjacent point to pickup the customer.

-If no taxi is free at that time, booking is rejected

Design modules for

1) Call taxi booking

Input 1:

Customer ID: 1

Pickup Point: A

Drop Point: B

Pickup Time: 9

Output 1:

Taxi can be allotted.

Taxi-1 is allotted

Input 2:

Customer ID: 2

Pickup Point: B

Drop Point: D

Pickup Time: 9

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Output 1:

Taxi can be allotted.

Taxi-2 is allotted

(Note: Since Taxi-1 would have completed its journey when second booking is done, so Taxi-2 from nearest point A which is free is allocated)

Input 3:

Customer ID: 3

Pickup Point: B

Drop Point: C

Pickup Time: 12

Output 1:

Taxi can be allotted.

Taxi-1 is allotted

2) Display the Taxi details

Taxi No: Total Earnings:

BookingID	CustomerID	From	To	PickupTime	DropTime	Amount
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Output:

Taxi-1 Total Earnings: Rs. 400

1	1	A	B	9	10	200
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3	3	B	C	12	13	200
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Taxi-2 Total Earnings: Rs. 350

2	2	B	D	9	11	350
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These were just sample inputs. It should work for any input that they give.

Those who finished both the modules within 3 hours and if it worked for all the inputs they give, those candidates were given extra modules to work with.